CYRILLE KENFACK

Software Engineer – Python & Java | Backend, Data/ML & Cloud

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PROFESSIONAL SUMMARY

Software Engineer with **5 years** of experience building production-grade backend services, REST APIs and data/ML platforms in **Python and Java**. Currently a consultant at **Max Digital Services**, productionizing **Machine Learning, Deep Learning and text-mining** algorithms on a cloud genomics and bioinformatics platform (**AWS, Kubernetes, FastAPI, Keycloak**). Previously R&D Software Engineer at **INRIA**, where I built a distributed testing framework for federated learning that cut validation time by **40%** with **90%+ test coverage**. Proven track record in microservices architecture, CI/CD pipelines, high availability and distributed systems.

CORE COMPETENCIES

Programming & Automation (5+ years)**: Python** (5 years) | **Java** (4 years) | **JavaScript** (3 years) | **Shell Scripting** (2 years)

Development Frameworks (4+ years)**: Spring Boot** (3 years) | **Django REST Framework** (3 years) | **FastAPI** | **Flask** | **React/Angular** (2 years)

Data, Machine Learning & MLOps (4+ years)**: Pandas/NumPy** (4 years) | **TensorFlow/PyTorch** (4 years) | **NLP & Text Mining (BERT, spaCy, Transformers)** (3 years) | **ML model productionization** | **Pytest**

DevOps Tools & Practices (4+ years)**: Git & Version Control** (5 years) | **Docker** (3 years) | **CI/CD (GitLab CI, Bitbucket)** (3 years) | **Microservices Architecture** (3 years) | **API Security** (3 years)

Cloud & Infrastructure (3+ years)**: AWS** | **Kubernetes** | **Keycloak (OAuth2/OIDC)** | **Google Cloud Platform (GCP)** | **Linux Administration** (3 years) | **Database Replication & Management** (2 years)

Database Technologies (4+ years)**: MySQL** (4 years) | **PostgreSQL** (3 years) | **MongoDB** (2 years) | **Database Optimization** (3 years)

PROFESSIONAL EXPERIENCE

Software Engineer – Python, Data & MLOps (Consultant) June 2026 – Present
Max Digital Services (IT consulting) – Client: global leader in in-vitro diagnostics *Rennes, France*

- Build Python data-analysis services (FastAPI, Pandas, NumPy) for a cloud SaaS genomics and bioinformatics platform running on AWS
- Productionize algorithms from data-science experiments (machine learning, deep learning, text mining) and data pipelines, from prototype to tested, containerized production service
- Ship robust, scalable solutions built to handle load: Docker Compose locally, Kubernetes on AWS in production, authentication delegated to Keycloak (OAuth2/OIDC)
- Equip the data-science and bioinformatics teams with tooling that eases the experiment-to-production transition (Poetry templates, argparse/python-fire CLIs)
- Keep the Python ecosystem state-of-the-art: version upgrades (pyenv, Poetry), Pytest, code quality (pylint, yapf, isort), Bitbucket CI
- Work in a Java Spring / React / Python monorepo with Kanban (Jira); documentation in Confluence, all deliverables in English

Technologies: Python 3, FastAPI, Pandas, NumPy, Poetry, Pytest, Docker Compose, Kubernetes, AWS, Keycloak, Bitbucket, Jira, Confluence

R&D Software Engineer September 2023 – May 2026
INRIA *Rennes, France*
- Developed and maintained a comprehensive testing framework for decentralized machine learning algorithms in cluster, production, and hybrid environments
- Implemented automated testing pipelines reducing deployment time by 40%
- Designed and developed high-performance Rust prototypes for federated learning algorithms
- Collaborated with international research teams on distributed systems architecture
- Optimized system performance and resource utilization in cloud environments
- Designed and implemented a discrete-event simulation environment to test large-scale distributed machine learning algorithms

Technologies: Python, TensorFlow, PyTorch, Rust, Linux, Distributed Systems

Backend Developer April 2023 – August 2023
moobifun *Yaoundé, Cameroon*
- Architected and developed scalable microservices using Spring Boot framework
- Implemented JPA ORM for efficient relational data management across multiple applications
- Secured APIs using Spring Security with OAuth2 and JWT authentication
- Established inter-service communication using OpenFeign for microservices architecture
- Developed comprehensive unit testing suites using JUnit achieving 85% code coverage
- Administered production databases and implemented MySQL slave replication for high availability

Technologies: Java, Spring Boot, Spring Security, OpenFeign, MySQL, MariaDB, Angular, Docker

Full-Stack Developer & IT Consultant October 2022 – March 2023
Connect Technology (Consultant at moobifun) *Yaoundé, Cameroon*
- Built secure microservices for financial transaction management using Spring Boot
- Designed and implemented user affiliation management API with role-based access control
- Developed real-time monitoring web application for system performance tracking
- Set up MySQL replication infrastructure for data redundancy and disaster recovery

Technologies: Java, Spring Boot, Spring Data, Spring Security, MySQL, Angular, Monitoring Tools

Software Engineer February 2022 – July 2022
AI Laboratory - ENSPY *Yaoundé, Cameroon*
- Developed an intelligent dialogue system using NLP for human-computer interaction
- Implemented machine learning models for diagnostic learning systems
- Analyzed user interaction data to optimize system performance and user experience
- Integrated TensorFlow and BERT models for natural language processing capabilities

Technologies: TensorFlow, PyAgrum, Spacy, BERT, React Native, Django REST, PostgreSQL

KEY PROJECTS

Flask Currency Converter *2025*
- Built production-ready Flask application with JWT authentication and Redis caching
- Implemented Docker containerization with multi-stage builds for optimized deployment
- Set up Celery task queue with Redis for asynchronous processing and background jobs
- Configured comprehensive testing suite with Pytest and coverage reporting

- Deployed using Gunicorn with horizontal scaling capabilities
- Integrated Swagger/OpenAPI documentation with 10+ documented endpoints
- **Technologies:** Python, Flask, Redis, PostgreSQL, Celery, Flask, Flask-JWT, Docker, Gunicorn

Agricultural Auction Platform *(January 2022)*
- Built a scalable web application for agricultural product auctions during COVID-19
- Implemented real-time bidding system with WebSocket connections
- Developed user management, favorites system, and event notification features
- **Technologies:** MongoDB, React.js, Express.js, Node.js, WebSocket

Healthcare Information System (iCare) *(June 2021)*
- Created mobile application providing healthcare facility information and remote diagnostics
- Implemented KNN machine learning model for symptom-based disease prediction
- Developed geolocation-based services for finding nearby pharmacies and hospitals
- **Technologies:** React Native, Django REST, PostgreSQL, Scikit-learn, KNN

Land Digitization System *(July 2021)*
- Developed GIS-based application for land parcel management and visualization
- Integrated spatial data management tools for polygon manipulation
- Optimized PostgreSQL with PostGIS for high-volume geospatial data processing
- **Technologies:** Django REST, PostgreSQL, PostGIS, GDAL, React Native

EDUCATION

École Nationale Supérieure Polytechnique de Yaoundé (ENSPY) *September 2017 – July 2022*
Master's Degree in Computer Engineering Yaoundé, Cameroon
Relevant Coursework: Data Structures, Software Methodology, Database Management, Data Mining, AI, Statistics, OOP

University of Yaounde I *September 2016 – July 2017*
Bachelor's Degree I in Mathematics Yaounde, Cameroon
Relevant Coursework: Differential and Integral Calculus, Linear Algebra, Probability, Statistics, Numerical Analysis

PROFESSIONAL CERTIFICATIONS

Programming & Development

- Introduction to HTML5 (University of Michigan)
- Initiation to programming in Java (EPFL)
- Introduction to Object Oriented Programming in Java (EPFL)
- Machine Learning (Stanford University)
- Python for Everybody Specialization (University of Michigan)
- Microsoft Python Development Professional Certificate (Microsoft)
- Neural Networks and Deep Learning (DeepLearning.AI)

Cloud & DevOps

- Google Cloud Fundamentals: Core Infrastructure (Google Cloud)
- Google Cloud Big Data and Machine Learning Fundamentals (Google Cloud)
- Docker Certified Associate (DCA) Specialization (LearnKarts, Coursera)

Database Management

- Database Design and Basic SQL in PostgreSQL (University of Michigan)
- Introduction to NoSQL Databases (IBM)

LEADERSHIP & ACHIEVEMENTS

Team Lead & Co-Founder, GiTech *June 2020 – Dec 2022*

- Lead a team of 7 developers in managing multiple software development projects
- Won first place in scientific competition with iCare mobile application at ENSPY
- Established agile development processes and code review standards
- Mentored junior developers in best practices and modern development frameworks

TECHNICAL PROFICIENCY

Expert Level (4-5 years)	**Advanced Level (2-3 years)**	**Intermediate Level (1-2 years)**
Python, Java, Linux, Git	Spring Boot, Django REST, FastAPI	Rust, MongoDB, Celery
Database Administration (MySQL, PostgreSQL)	Pandas, NumPy, TensorFlow, PyTorch	AWS, Kubernetes, Keycloak
System Architecture	Docker, CI/CD, Microservices	NLP / Text Mining (BERT, spaCy)
REST API Development	API Security (OAuth2, JWT), Redis	Flask, Cloud (GCP)

LANGUAGES

French: Native **English:** Fluent (Professional Working Proficiency)